MILITARY SPECIFICATION SHEET

ELECTRON TUBE, PHOTOMULTIPLIER

TYPE 931VA

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: 9-stage, side-on, S4 spectral response

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ebb	Eb-Edy9	Light source	Light	Ib	TA	Rp	Cathode area Lighted	Shock	Alt
Unit:	Vdc	Vdc	---	ℓm	mAdc	°C	Ohms	sq cm	G	ft
Maximum:	1,250 (Notes 1 and 19)	250 (Notes 2 and 19)	---	---	1.0 (Note 10b)	75	Note 3	---	200	40,000
TEST CONDITIONS:	1,000	100	LS1	10^{-5}	---	---	10,000	0.8	---	---

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
1031	Low-frequency (25 Hz) vibration	4	No voltages; t = 60 minutes	---	---	---	---	---	---
1331	Direct-interelectrode capacitance (1)	8		---	---	C	2.0	6.0	pF
1331	Direct-interelectrode capacitance (2)	9		---	---	C	3.5	9.5	pF
(D) ---	Low-pressure breakdown	12, 13, 14	Ebb = 1,250 Vdc; Eb to Eby9 = 125 Vdc; voltages for all other stages = 125 Vdc per stage; Rp = 1 Meg; t = 120	---	---	---	---	---	---
5303	Spectral response identification	-	S4	---	---	---	---	---	---
(D) 1216	Base material insulating quality	-		---	---	---	---	---	---
	Quality conformance inspection, part 1	18							
---	Cathode current	6, 8	Light = 0.01 ℓm	---	---	Ik	0.06	---	μAdc
5301	Dark current (1)	-	Light = 0	---	---	Lik	---	5.0	μAdc
5301	Dark current (2)	5, 15		---	---	Lib	---	0.05	μAdc
5301	Anode current	5		---	---	Ib	45	---	μAdc
---	Amplification	-	Amplification = $\frac{Ib(\lambda/\ell m)}{Ik(\lambda/\ell m)}$	---	---	Mu	75,000	---	---
5301	Dark current (3)	7	Light = 0	---	---	Lib	---	5.0	μAdc
	Quality conformance inspection, part 2								
1101	Secureness of base, cap, or insert	11		---	---	---	---	---	---
1041	Shock	16, 17	200 G ± 10%; no voltages applied	---	---	---	---	---	---
(D) 1111	Base pin solder depth	-		---	---	---	---	---	---
(D) 1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	10	Group C	---	---	---	---	---	---
---	Life-test end point: Amplification	-		---	---	Mu	35,000	---	---

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NOTES:

1. Overall anode to cathode dc supply voltage.

2. Anode to 9th dynode dc supply voltage.

3. An anode load resistance of at least 10,000 ohms is recommended for a protective resistance.

4. Vibration test is to be performed prior to other qualification inspection tests.

5. Adjust tube position to give maximum sensitivity. The light during test, except during life test, shall be incident upon the aperture approximately 0.8 inch x 0.2 inch (20.32 mm x 5.08 mm) located to center of the cathode.

6. Apply 100 volts between the cathode and all other pins tied together.

7. Apply Ebb according to the range of Ib in the following table:

	Ib	Ebb
Range	45 to 63	1,600
	63 to 88	1,550
	88 to 120	1,500
	120 to 162	1,450
	162 to 214	1,400
	214 to 284	1,350
	284 to 374	1,300
	374 and greater	1,250

8. Capacitance between anode and dy9, all other elements grounded.

9. Capacitance between anode and all other elements.

10. Life test.

    (a) Life test shall be made during a cycle of 6 hours on and 18 hours off. The life test duration of 500 hours shall consist of 21 cycles. The luminous flux at the cathode shall be adjusted initially and at the end of 15 minutes operation daily to give an anode current of 2.5mAdc. Color temperature shall be greater than 2,000°K.

    (b) Longer tube life may be expected when operating conditions require lower anode current.

11. Omit immersion in water and test dry. Apply torque test with 20.0 pound-inches minimum.

12. Test circuit and test equipment requirements.

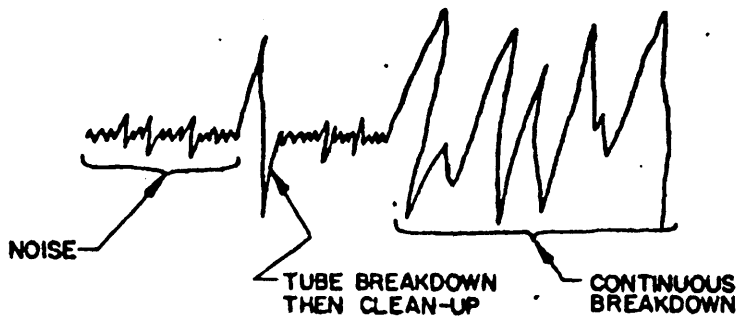
    A bleeder network to provide equal voltage per stage; 8-megohm series resistance for cathode, and all dynodes excluding the last dynode (9th); 1-megohm series resistance for last dynode (9th) and anode; a regulated (0.1 percent) power supply with low-ripple (0.05 volts peak-to-peak maximum) as a voltage source; a sensitive dc microammeter to determine anode current; an oscilloscope, with a low-capacity divider, is connected across the anode load resistance (1-megohm) to monitor noise and breakdown; base pins of photomultiplier under test connected so that the degradation of breakdown characteristic is minimized; a vacuum pump, low-pressure chamber and manometer; a light source with an adjustable iris and sand-blasted glass.

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NOTES: - Continued

Test method.

Mask tube with light-proof material. The light-source should be located so that light from it is striking the cathode of the photomultiplier. (Light from the light source should be the only light striking the cathode.) The opening of the iris is varied to obtain approximately 10 microamperes of anode current. Observe the tube noise signal as displayed by the oscilloscope. Reduce pressure of vacuum chamber to 130 millimeters of mercury. The tube should be subjected to this pressure for a period of 2 minutes. During pressure reduction, the tube noise signal, as displayed by the oscilloscope, should be observed to determine tube breakdown. When breakdown occurs, the signal being monitored by the oscilloscope will resemble the following:



When continuous breakdown occurs, a change in anode current will result.

13. Vary light to obtain approximately 10 microamperes of anode current.

14. The tube shall operate without indication of breakdown in an evacuated chamber in which the pressure does not exceed 130 mmHg absolute.

15. With 10^{-5} lumen incident on the cathode, vary the overall voltage (Ebb) to obtain an anode current of 200 μAdc. Exclude light and read the dark current.

16. The clamp shall be as specified in Drawing 216-JAN.

17. After the shock test is performed, the tube shall meet the requirements of all the tests specified in quality conformance inspection, part 1.

18. All quality conformance inspection, part 1, tests to be performed at the conclusion of the holding period.

19. An ac supply voltage may be used having a peak value equal to this maximum value.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11
DSA - ES

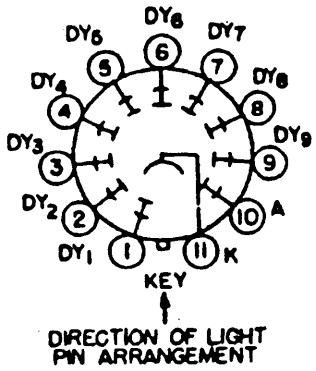
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH

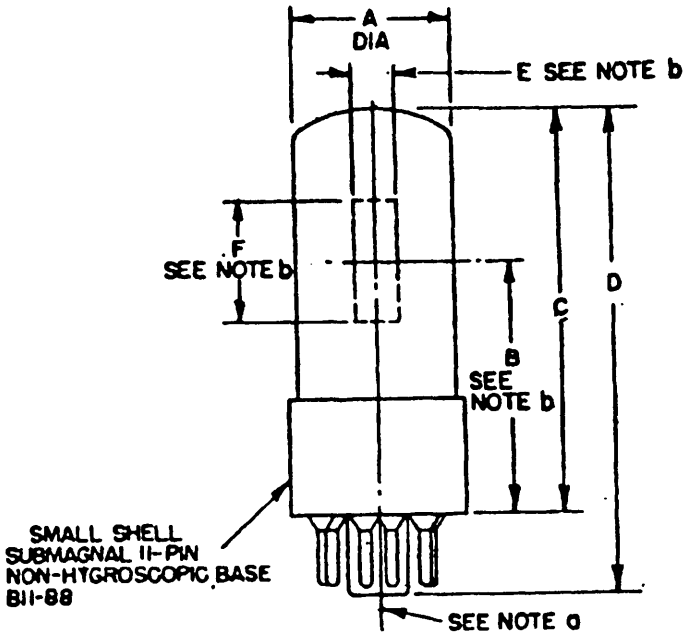
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2555)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
' Qualification inspection		
C		3.125(79.38)
E	.312 (7.92)	
F	.937(23.80)	
Quality conformance inspection, part 2		
A		1.187(30.15)
B	1.843(46.81)	2.031(51.59)
D		3.687(93.65)

NOTES:
a. Centerline of bulb shall not deviate more than 2° in any direction from the perpendicular erected at center of bottom of base.
b. Dimensions B, E, and F refer to the usable cathode area.

FIGURE 1. Outline drawing of electron tube type 931VA.

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